CONG WANG

Ph.D. Candidate ⋄ Microelectronic Thrust
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RESEARCH INTERESTS

Electronic Design Automation (EDA), **ML Accelerators**, **Domain Specific Architecture**

EDUCATION

The Hong Kong University of Science and Technology (Guangzhou) *Sept. 2023 - Present*
Microelectronic Thrust
Supervisors: Prof. Shanshi Huang (Ph.D., Georgia Institute of Technology)
Prof. Wei Zhang (Ph.D., Princeton University; IEEE Fellow) Ph.D. Candidate

Southern University of Science and Technology *Sept. 2021 - Jul. 2023*
School of Microelectronic | *Supervisor: Prof. Quan Chen (Ph.D., The University of Hong Kong)* M.Eng

Zhengzhou University *Sept. 2017 - Jul. 2021*
School of Information Engineering B.Eng

SELECTED PUBLICATIONS

[1] **Cong Wang**, Liangcheng Zhao, Zexin Fu, Jing Zhang, Xunqin Lai, Jiayi Huang, and Shanshi Huang. "A Scalable Cluster-based Heterogeneous CIM Chiplet Architecture for Vision Transformers." *IEEE Transactions on Computers*, under review.

[2] **Cong Wang**, Zexin Fu, Jiayi Huang, and Shanshi Huang. "Hemlet: A Heterogeneous Compute-in-Memory Chiplet Architecture for Vision Transformers with Group-Level Parallelism." *The 44th IEEE International Conference on Computer Design (ICCD)*, 2026.

[3] Ruihao He, **Cong Wang**, Liangcheng Zhao, Xipeng Lin, Shaoxuan Li, and Hongwu Jiang. "Quantifying Scalability-Performance Trade-offs: A Design Space Exploration of Compute-in-Memory based Ising Machines." *ACM Transactions on Design Automation of Electronic Systems*, 2026.

[4] Shaoxuan Li, Zikun Wei, Ruihao He, **Cong Wang**, Ye Gu, Tianchu Dong, Yucheng Cai, and Hongwu Jiang. "TOKCIM: A 65nm All-Digital Compute-in-Memory Transformer Accelerator with Advanced Weight Sparsity Handling and SoftMax Acceleration." *IEEE Custom Integrated Circuits Conference (CICC)*, 2026.

[5] Yihang Zuo, Zexin Fu, **Cong Wang**, Yucao Wu, Jiayi Huang, and Yuzhe Ma. "Harmony: A Hardware-Mapping Co-Exploration Framework for Hybrid CIM-based Vision Transformer Accelerator." *International Symposium on Quality Electronic Design (ISQED)*, 2026. **Best Paper Award.**

[6] Ruihao He, **Cong Wang**, Liangcheng Zhao, Zefeng Xu, and Hongwu Jiang. "Electronic-Photonic Hybrid CIM-based Ising Machine with Interaction-Aware Update Strategy for Combinational Optimization Problems." *IEEE International Symposium on Circuits and Systems (ISCAS)*, 2026.

[7] Zhengxi Yan, **Cong Wang**, Xipeng Lin, Jing Zhang, and Hongwu Jiang. "CIM-SIFT: An Efficient Compute-in-Memory based Accelerator for Real-time SIFT Algorithm." *IEEE International Symposium on Circuits and Systems (ISCAS)*, 2026.

[8] **Cong Wang**, Zeming Chen, and Shanshi Huang. "MICSim: A Modular Pre-Circuit Simulator for Mixed-Signal Compute-in-Memory Accelerators in CNNs and Transformers." *Integration, the VLSI Journal*, 2025.

[9] Ruihao He, **Cong Wang**, Xipeng Lin, Shaoxuan Li, and Hongwu Jiang. "RCIM: A Reconfigurable Compute-in-Memory Architecture for Artificial, Spiking and Hybrid Neural Networks." *IEEE International Conference on Integrated Circuits, Technologies and Applications (ICTA)*, 2025.

[10] Xipeng Lin, **Cong Wang**, Shanshi Huang, and Hongwu Jiang. "An Efficient Compute-in-Memory based Accelerator for Point-based Point Cloud Neural Networks." *ACM/IEEE Design Automation Conference (DAC)*, 2025.

[11] **Cong Wang**, Zeming Chen, and Shanshi Huang. "MICSim: A Modular Simulator for Mixed-signal Compute-in-Memory based AI Accelerator." *Asia and South Pacific Design Automation Conference (ASP-DAC)*, 2025.

[12] **Cong Wang**, Dongen Yang, Jinming Lyu, Yong Dai, Cheng Zhuo, and Quan Chen. "On Model Order Reduction and Exponential Integrator for Transient Circuit Simulation." *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems*, 2024.

[13] **Cong Wang**, Dongen Yang, and Quan Chen. "EI-MOR: A Hybrid Exponential Integrator and Model Order Reduction Approach for Transient Power/Ground Network Analysis." *IEEE/ACM International Conference on Computer-Aided Design (ICCAD)*, 2022.

RESEARCH AND PROJECT EXPERIENCE

MICSim: Open-source CIM Accelerator Simulator *Sept. 2023 - Jul. 2024*

- First contributor; developed a modular simulator for mixed-signal CIM accelerators in CNNs and Transformers.
- Unified model-accuracy and hardware-performance evaluation; GitHub: MICSim_V1.0 has 39 stars.

Tiny TPU: MICS6000U Course Project *Fall 2024*

- Teaching assistant and lead contributor; implemented a compact TPU in SystemVerilog and authored course tutorials and cocotb testbenches.

Transient Circuit Simulation Acceleration *Jan. 2022 - Jul. 2023*

- Core contributor to an NSFC project; developed exponential-integrator methods to accelerate large-scale transient circuit simulation.

Model Order Reduction for Circuit Simulation *Jul. 2021 - Aug. 2022*

- Primary contributor to a Huawei HiSilicon collaboration; developed industrial-grade C++ programs for model order reduction methods.

HONORS AND AWARDS

- HKUST(GZ) Research Postgraduate *2023-present*
- SUSTech Outstanding Graduate *2023*
- Best Paper Nomination, National Graduate Forum on Microelectronics *2023*
- Third Place, Integrated Circuit EDA Elite Challenge *2023*
- SUSTech Postgraduate Scholarship *2021-2023*
- Zhengzhou University Third Prize Scholarship *2021*
- National Encouragement Scholarship *2019, 2020*
- Merit Student of Zhengzhou University *2019, 2020*
- First Prize, Zhengzhou University National Mathematics Competition *2020*
- Second Prize, Henan Province National Mathematics Competition *2020*
- First Place, Zhengzhou University "Qiushi Cup" Mathematics Competition *2020*
- First Prize, "Challenge Cup" National College Student Business Plan Competition *2019*
- Excellent Student Cadre *2019*
- Advanced Individual of Social Work *2018*
- Excellent Student Volunteer *2018*

TEACHING

MICS6000U ML Accelerator, Teaching Assistant *Fall 2024*
UCMP6010 Cross-disciplinary Research Methods, Teaching Assistant *Spring 2025*

SKILLS

Programming: Python (NumPy, SciPy), C/C++, MATLAB.
Hardware & Verification: Verilog/SystemVerilog, cocotb; **AI Framework**: PyTorch.
Software & Tools: LaTeX, Markdown, MS Office; **Language**: Mandarin (Native), English (IELTS 7.0, CET-6).